

Yu Xiang

CONTACT INFORMATION	Assistant Professor Director, Intelligent Robotics and Vision Lab Department of Computer Science The University of Texas at Dallas 800 W. Campbell Road, Richardson, TX 75080	yu.xiang@utdallas.edu https://labs.utdallas.edu/irvl/ http://yuxng.github.io/ Google Scholar GitHub
RESEARCH INTERESTS	Robotics, Robot Learning, Robot Manipulation, Computer Vision	
EXPERIENCE	The University of Texas at Dallas , Richardson, Texas, USA <i>Assistant Professor</i>	Aug 2021 – present
	NVIDIA Research , Seattle, Washington, USA <i>Senior Research Scientist</i>	Jun 2018 – Jul 2021
	NVIDIA Research , Seattle, Washington, USA <i>Postdoctoral Researcher</i>	Jan 2018 – May 2018
	University of Washington , Seattle, Washington, USA <i>Postdoctoral Researcher</i> • Advisor: Prof. Dieter Fox	Aug 2016 – Dec 2017
	Stanford University , Stanford, California, USA <i>Postdoctoral Researcher</i> • Advisor: Prof. Silvio Savarese	Jan 2016 – Jul 2016
	Stanford University , Stanford, California, USA <i>Visiting Student Researcher</i> • Advisor: Prof. Silvio Savarese	Sep 2013 – Dec 2015
	NEC Laboratories America, Inc. , Cupertino, California, USA <i>Summer Research Intern</i> <i>Summer Research Intern</i> • Department: Media Analytics	Jun 2015 – Sep 2015 May 2014 – Aug 2014
EDUCATION	University of Michigan , Ann Arbor, Michigan, USA Ph.D. in Electrical Engineering: Systems Dissertation: 3D Object Representations for Recognition Advisor: Prof. Silvio Savarese	Sep 2010 – Dec 2015
	Fudan University , Shanghai, China M.S. in Computer Science Dissertation: Graphic Models for Semantic Context Modeling in Automatic Image Annotation Advisor: Prof. Xiangdong Zhou	Sep 2007 – Jul 2010
	Fudan University , Shanghai, China B.S. in Computer Science	Sep 2003 – Jul 2007
AWARDS AND HONORS	IEEE Senior Member	2026
	NVIDIA Academic Grant Program Award	2024
	Sony Research Award	2022
	One of 12 ECCV Best Papers Selected for an IJCV Special Issue	2018
	University of Washington CSE Postdoc Research Award	2016
	ICCV Doctoral Consortium Travel Award	2015
	Outstanding Master's Thesis Award, Shanghai	2012
RESEARCH FUNDING	1. FRR: Learning Object-Centric Representations from Human Demonstrations for Robot Manipulation, NSF, PI, \$399,158, 09/01/2025 – 08/31/2028	

2. NSF CISE REU Student Funding, PI, \$10,000, 2026 (associated with NSF Award #2520553)
3. CIRC: Dev: SMILE: Scan to Multi-sensorial Interactive Learning Environment, NSF, co-PI, \$750,000 total award, 07/01/2024 – 06/30/2026
4. Frontier Mining Project, PI, \$50,000, 09/01/2025 – 03/31/2026
5. XPeng Gift, PI, \$70,000, 09/01/2024 – 08/31/2025
6. Sony Research Award, PI, \$100,000, 09/01/2023 – 08/31/2024
7. Perceptually-enabled Task Guidance, DARPA, co-PI, \$3,700,000 total award, 10/01/2021 – 09/30/2025

PUBLICATIONS

1. **NAC: Neural Action Codec for Vision-Language-Action Models**
Ahad Jawaaid, **Yu Xiang**
In Conference on Robot Learning (CoRL), 2026.
2. **Cross-Embodiment Robot Manipulation via a Unified Hand Action Space**
Luis Felipe Casas, Robert Teal, Keval Shah, Abhijit Tadepalli, Wanxin Jin, **Yu Xiang**
In 4th Workshop on Dexterous Manipulation at Robotics: Science and Systems (RSS), 2026.
3. **VLA-REPLICA: A Low-Cost, Reproducible Benchmark for Real-World Evaluation of Vision-Language-Action Models**
Alex S. Huang*, Jiahui Zhang*, Shiqing Tang, **Yu Xiang** (*equal contribution)
In Data-Centric Robotics Workshop at Robotics: Science and Systems (RSS), 2026.
4. **Build Once, Monitor Continuously: Persistent Semantic Mapping via Autonomous Exploration and Open-Vocabulary Object Updates**
Sai Haneesh Allu, Itay Kadosh, Tyler Summers, **Yu Xiang**
In WORLDS: World Models and Spatial Intelligence for Physical AI Workshop (IROS), 2026.
5. **Continual Distillation Learning for Rehearsal-Free Class-Incremental Learning via Decoupled Prompting**
Qifan Zhang, Yunhui Guo, **Yu Xiang**
In Conference on Lifelong Learning Agents (CoLLAs), 2026.
6. **From Local Matches to Global Masks: Template-Guided Instance Detection and Segmentation in Open-World Scenes**
Qifan Zhang, Sai Haneesh Allu, Jikai Wang, Yangxiao Lu, **Yu Xiang**
In *Robotics: Science and Systems (RSS)*, 2026.
7. **ROBOMETER: Scaling General-Purpose Robotic Reward Models via Trajectory Comparisons**
Anthony Liang, Yigit Korkmaz, Jiahui Zhang, Minyoung Hwang, Abrar Anwar, Sidhant Kaushik, Aditya Shah, Alex S. Huang, Luke Zettlemoyer, Dieter Fox, **Yu Xiang**, Anqi Li, Andreea Bobu, Abhishek Gupta, Stephen Tu, Erdem Biyik, Jesse Zhang
In *Robotics: Science and Systems (RSS)*, 2026.
8. **Motion Planning with Precedence Specifications via Augmented Graphs of Convex Sets**
Shilin You, Gael Luna, Juned Shaikh, David Gostin, **Yu Xiang**, Justin Koeln, Tyler Summers
In *International Conference on Robotics and Automation (ICRA)*, 2026.
9. **HO-Cap: A Capture System and Dataset for 3D Reconstruction and Pose Tracking of Hand-Object Interaction**
Jikai Wang, Qifan Zhang, Yu-Wei Chao, Bowen Wen, Xiaohu Guo, **Yu Xiang**
In *NeurIPS 2025 Track on Datasets and Benchmarks (NeurIPS)*, 2025.
10. **Adapting Pre-Trained Vision Models for Novel Instance Detection and Segmentation**
Yangxiao Lu, Jishnu Jaykumar P, Yunhui Guo, Nicholas Ruoizzi, **Yu Xiang**
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.
11. **RobotFingerPrint: Unified Gripper Coordinate Space for Multi-Gripper Grasp Synthesis and Transfer**
Ninad Khargonkar, Luis Felipe Casas, Balakrishnan Prabhakaran, **Yu Xiang**
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2025.

12. **V-HOP: Visuo-Haptic 6D Object Pose Tracking**
Hongyu Li, Mingxi Jia, Tuluhan Akbulut, **Yu Xiang**, George Konidakis, Srinath Sridhar
In *Robotics: Science and Systems (RSS)*, 2025.
13. **CaptainCook4D: A Dataset for Understanding Errors in Procedural Activities**
Rohith Peddi, Shivvrat Arya, Bharath Challa, Likhitha Pallapothula, Akshay Vyas, Bhavya Gouripeddi, Qifan Zhang, Jikai Wang, Vasundhara Komaragiri, Eric Ragan, Nicholas Ruoizzi, **Yu Xiang**, Vibhav Gogate
In *NeurIPS 2024 Track on Datasets and Benchmarks (NeurIPS)*, 2024.
14. **MultiGripperGrasp: A Dataset for Robotic Grasping from Parallel Jaw Grippers to Dexterous Hands**
Luis Felipe Casas*, Ninad Khargonkar*, Balakrishnan Prabhakaran, **Yu Xiang** (*equal contribution)
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
15. **Grasping Trajectory Optimization with Point Clouds**
Yu Xiang, Sai Haneesh Allu, Rohith Peddi, Tyler Summers, Vibhav Gogate
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
16. **PROTO-CLIP: Vision-Language Prototypical Network for Few-Shot Learning**
Jishnu Jaykumar P, Kamalesh Palanisamy, Yu-Wei Chao, Xinya Du, **Yu Xiang**
In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, 2024.
17. **Segment Every Out-of-Distribution Object**
Wenjie Zhao, Jia Li, Xin Dong, **Yu Xiang**, Yunhui Guo
In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2024.
18. **SceneReplica: Benchmarking Real-World Robot Manipulation by Creating Reproducible Scenes**
Ninad Khargonkar, Sai Haneesh Allu, Yangxiao Lu, Jishnu Jaykumar P, Balakrishnan Prabhakaran, **Yu Xiang**
In *International Conference on Robotics and Automation (ICRA)*, 2024.
19. **RISeg: Robot Interactive Object Segmentation via Body Frame-Invariant Features**
Howard H. Qian, Yangxiao Lu, Kejia Ren, Gaotian Wang, Ninad Khargonkar, **Yu Xiang**, Kaiyu Hang
In *International Conference on Robotics and Automation (ICRA)*, 2024.
20. **Mean Shift Mask Transformer for Unseen Object Instance Segmentation**
Yangxiao Lu, Yuqiao Chen, Nicholas Ruoizzi, **Yu Xiang**
In *International Conference on Robotics and Automation (ICRA)*, 2024.
21. **Deep Dependency Networks for Multi-Label Classification**
Shivvrat Arya, **Yu Xiang**, Vibhav Gogate
In *International Conference on Artificial Intelligence and Statistics (AISTATS)*, 2024.
22. **Self-Supervised Unseen Object Instance Segmentation via Long-Term Robot Interaction**
Yangxiao Lu, Ninad Khargonkar, Zesheng Xu, Charles Averill, Kamalesh Palanisamy, Kaiyu Hang, Yunhui Guo, Nicholas Ruoizzi, **Yu Xiang**
In *Robotics: Science and Systems (RSS)*, 2023.
23. **FewSOL: A Dataset for Few-Shot Object Learning in Robotic Environments**
Jishnu Jaykumar P, Yu-Wei Chao, **Yu Xiang**
In *International Conference on Robotics and Automation (ICRA)*, 2023.
24. **NeuralGrasps: Learning Implicit Representations for Grasps of Multiple Robotic Hands**
Ninad Khargonkar, Neil Song, Zesheng Xu, Balakrishnan Prabhakaran, **Yu Xiang**
In *Conference on Robot Learning (CoRL)*, 2022.
25. **Few-shot Single-view 3D Reconstruction with Memory Prior Contrastive Network**
Zhen Xing, Yijiang Chen, Zhixin Ling, Xiangdong Zhou, **Yu Xiang**
In *European Conference on Computer Vision (ECCV)*, 2022.
26. **TALISMAN: Targeted Active Learning for Object Detection with Rare Classes and Slices using Submodular Mutual Information**
Suraj Kothawade, Saikat Ghosh, Sumit Shekar, **Yu Xiang**, Rishabh Iyer
In *European Conference on Computer Vision (ECCV)*, 2022.

27. **HandoverSim: A Simulation Framework and Benchmark for Human-To-Robot Object Handovers**
 Yu-Wei Chao, Chris Paxton, **Yu Xiang**, Wei Yang, Balakumar Sundaralingam, Tao Chen, Adithyavairavan Murali, Maya Cakmak and Dieter Fox
 In *International Conference on Robotics and Automation (ICRA)*, 2022.
28. **Hierarchical Policies for Cluttered-Scene Grasping with Latent Plans**
 Lirui Wang, Xiangyun Meng, **Yu Xiang** and Dieter Fox
 In *IEEE Robotics and Automation Letters (RA-L)*, 2022.
29. **iCaps: Iterative Category-level Object Pose and Shape Estimation**
 Xinke Deng*, Junyi Geng*, Timothy Bretl, **Yu Xiang** and Dieter Fox (*equal contribution)
 In *IEEE Robotics and Automation Letters (RA-L)*, 2022.
30. **RICE: Refining Instance Masks in Cluttered Environments with Graph Neural Networks**
 Christopher Xie, Arsalan Mousavian, **Yu Xiang** and Dieter Fox
 In *Conference on Robot Learning (CoRL)*, 2021.
31. **Goal-Auxiliary Actor-Critic for 6D Robotic Grasping with Point Clouds**
 Lirui Wang, **Yu Xiang** and Dieter Fox
 In *Conference on Robot Learning (CoRL)*, 2021.
32. **DexYCB: A Benchmark for Capturing Hand Grasping of Objects**
 Yu-Wei Chao, Wei Yang, **Yu Xiang**, Pavlo Molchanov, Ankur Handa, Jonathan Tremblay, Yashraj Narang, Karl Van Wyk, Umar Iqbal, Stan Birchfield, Jan Kautz and Dieter Fox
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
33. **RGB-D Local Implicit Function for Depth Completion of Transparent Objects**
 Luyang Zhu, Arsalan Mousavian, **Yu Xiang**, Hammad Mazhar, Jozef van Eenbergen, Shoubhik Debnath and Dieter Fox
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2021.
34. **Learning Composable Behavior Embeddings for Long-horizon Visual Navigation**
 Xiangyun Meng, **Yu Xiang** and Dieter Fox
 In *IEEE Robotics and Automation Letters (RA-L)*, 2021.
35. **Unseen Object Instance Segmentation for Robotic Environments**
 Christopher Xie, **Yu Xiang**, Arsalan Mousavian and Dieter Fox
 In *IEEE Transactions on Robotics (T-RO)*, 2021.
36. **PoseRBPF: A Rao-Blackwellized Particle Filter for 6D Object Pose Tracking**
 Xinke Deng, Arsalan Mousavian, **Yu Xiang**, Fei Xia, Timothy Bretl and Dieter Fox
 In *IEEE Transactions on Robotics (T-RO)*, 2021.
37. **Learning RGB-D Feature Embeddings for Unseen Object Instance Segmentation**
Yu Xiang, Christopher Xie, Arsalan Mousavian and Dieter Fox
 In *Conference on Robot Learning (CoRL)*, 2020.
38. **Manipulation Trajectory Optimization with Online Grasp Synthesis and Selection**
 Lirui Wang, **Yu Xiang** and Dieter Fox
 In *Robotics: Science and Systems (RSS)*, 2020.
39. **LatentFusion: End-to-End Differentiable Reconstruction and Rendering for Unseen Object Pose Estimation**
 Keunhong Park, Arsalan Mousavian, **Yu Xiang** and Dieter Fox
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2020.
40. **Scaling Local Control to Large-Scale Topological Navigation**
 Xiangyun Meng, Nathan Ratliff, **Yu Xiang** and Dieter Fox
 In *International Conference on Robotics and Automation (ICRA)*, 2020.
41. **Self-supervised 6D Object Pose Estimation for Robot Manipulation**
 Xinke Deng, **Yu Xiang**, Arsalan Mousavian, Clemens Eppner, Timothy Bretl and Dieter Fox
 In *International Conference on Robotics and Automation (ICRA)*, 2020.

42. **The Best of Both Modes: Separately Leveraging RGB and Depth for Unseen Object Instance Segmentation**
Christopher Xie, **Yu Xiang**, Arsalan Mousavian and Dieter Fox
In *Conference on Robot Learning (CoRL)*, 2019.
43. **PoseRBPF: A Rao-Blackwellized Particle Filter for 6D Object Pose Tracking**
Xinke Deng, Arsalan Mousavian, **Yu Xiang**, Fei Xia, Timothy Bretl and Dieter Fox
In *Robotics: Science and Systems (RSS)*, 2019.
44. **Object Discovery in Videos as Foreground Motion Clustering**
Christopher Xie, **Yu Xiang**, Dieter Fox and Zaid Harchaoui
In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019.
45. **Neural Autonomous Navigation with Riemannian Motion Policy**
Xiangyun Meng, Nathan Ratliff, **Yu Xiang** and Dieter Fox
In *International Conference on Robotics and Automation (ICRA)*, 2019.
46. **Deep Object Pose Estimation for Semantic Robotic Grasping of Household Objects**
Jonathan Tremblay, Thang To, Balakumar Sundaralingam, **Yu Xiang**, Dieter Fox and Stan Birchfield
In *Conference on Robot Learning (CoRL)*, 2018.
47. **DeepIM: Deep Iterative Matching for 6D Pose Estimation**
Yi Li, Gu Wang, Xiangyang Ji, **Yu Xiang** and Dieter Fox
In *European Conference on Computer Vision (ECCV)*, 2018 **-Oral**.
48. **PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes**
Yu Xiang, Tanner Schmidt, Venkatraman Narayanan and Dieter Fox
In *Robotics: Science and Systems (RSS)*, 2018.
49. **Recurrent Autoregressive Networks for Online Multi-Object Tracking**
Kuan Fang, **Yu Xiang**, Xiaocheng Li and Silvio Savarese
In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, 2018.
50. **DA-RNN: Semantic Mapping with Data Associated Recurrent Neural Networks**
Yu Xiang and Dieter Fox
In *Robotics: Science and Systems (RSS)*, 2017.
51. **Subcategory-aware Convolutional Neural Networks for Object Proposals and Detection**
Yu Xiang, Wongun Choi, Yuanqing Lin and Silvio Savarese
In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pp. 924–933, 2017.
52. **Anticipating Accidents in Dashcam Videos**
Fu-Hsiang Chan, Yu-Ting Chen, **Yu Xiang** and Min Sun
In *Asian Conference on Computer Vision (ACCV)*, pp. 136–153, 2016 **-Oral**.
53. **ObjectNet3D: A Large Scale Database for 3D Object Recognition**
Yu Xiang, Wonhui Kim, Wei Chen, Jingwei Ji, Christopher Choy, Hao Su, Roozbeh Mottaghi, Leonidas Guibas and Silvio Savarese
In *European Conference on Computer Vision (ECCV)*, pp. 160–176, 2016 **-Spotlight Oral**.
54. **Pose Estimation Errors, the Ultimate Diagnosis**
Carolina Redondo-Cabrera, Roberto López-Sastre, **Yu Xiang**, Tinne Tuytelaars and Silvio Savarese
In *European Conference on Computer Vision (ECCV)*, pp. 118–134, 2016.
55. **Deep Metric Learning via Lifted Structured Feature Embedding**
Hyun Oh Song, **Yu Xiang**, Stefanie Jegelka and Silvio Savarese
In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4004–4012, 2016 **-Spotlight Oral**.
56. **Learning to Track: Online Multi-Object Tracking by Decision Making**
Yu Xiang, Alexandre Alahi and Silvio Savarese
In *International Conference on Computer Vision (ICCV)*, pp. 4705–4713, 2015 **-Oral**.

57. **Data-Driven 3D Voxel Patterns for Object Category Recognition**
Yu Xiang, Wongun Choi, Yuanqing Lin and Silvio Savarese
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1903–1911, 2015 **-Oral**.
58. **A Coarse-to-Fine Model for 3D Pose Estimation and Sub-category Recognition**
 Roozbeh Mottaghi, **Yu Xiang** and Silvio Savarese
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 418–426, 2015.
59. **Monocular Multiview Object Tracking with 3D Aspect Parts**
Yu Xiang*, Changkyu Song*, Roozbeh Mottaghi and Silvio Savarese (*equal contribution)
 In *European Conference on Computer Vision (ECCV)*, pp. 220–235, 2014.
60. **Beyond PASCAL: A Benchmark for 3D Object Detection in the Wild**
Yu Xiang, Roozbeh Mottaghi and Silvio Savarese
 In *IEEE Winter Conference on Applications of Computer Vision (WACV)*, pp. 75–82, 2014.
61. **Object Detection by 3D Aspectlets and Occlusion Reasoning**
Yu Xiang and Silvio Savarese
 In *IEEE Workshop on 3D Representation and Recognition (3dRR)*, pp. 530–537, 2013.
62. **Object Co-detection**
 Sid Yingze Bao, **Yu Xiang** and Silvio Savarese
 In *European Conference on Computer Vision (ECCV)*, vol. 7572, pp. 86–101, 2014.
63. **Estimating the Aspect Layout of Object Categories**
Yu Xiang and Silvio Savarese
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3410–3417, 2012.
64. **Semantic Context Modeling with Maximal Margin Conditional Random Fields for Automatic Image Annotation**
Yu Xiang, Xiangdong Zhou, Zuotao Liu, Tat-Seng Chua and Chong-Wah Ngo
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 3368–3375, 2010.
65. **Learning Contextual Metrics for Automatic Image Annotation**
 Zuotao Liu, Xiangdong Zhou, **Yu Xiang** and Yan-Tao Zheng
 In *Advances in Multimedia Information Processing - PCM*, vol. 6297, pp. 124–135, 2010.
66. **A Revisit of Generative Model for Automatic Image Annotation using Markov Random Fields**
Yu Xiang, Xiangdong Zhou, Zuotao Liu, Tat-Seng Chua and Chong-Wah Ngo
 In *IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 1153–1160, 2009.
67. **Adaptive Model for Web Image Semantic Automatic Image Annotation**
 Hongtao Xu, Xiangdong Zhou, **Yu Xiang** and Baile Shi
 In *Journal of Software (in Chinese)*, vol. 21, no. 9, pp. 2183–2195, 2009.
68. **Exploiting Flickr’s Related Tags for Semantic Annotation of Web Images**
 Hongtao Xu, Xiangdong Zhou, **Yu Xiang** and Baile Shi
 In *ACM International Conference on Image and Video Retrieval (CIVR)*, no. 46, 2009.
69. **Automatic Web Image Annotation via Web-Scale Image Semantic Space Learning**
 Hongtao Xu, Xiangdong Zhou, Lan Lin, **Yu Xiang** and Baile Shi
 In *Advances in Data and Web Management*, vol. 5446, pp. 211–222, 2009.

TEACHING

CS 6341 Robotics, University of Texas at Dallas, Dallas, Texas, USA

Fall 2026

CS 4391 Introduction to Computer Vision, University of Texas at Dallas, Dallas, Texas, USA
 Spring 2026

CS 6341 Robotics, University of Texas at Dallas, Dallas, Texas, USA

Fall 2025

CS 4391 Introduction to Computer Vision, University of Texas at Dallas, Dallas, Texas, USA
 Spring 2025

CS 6301 Special Topics in Computer Science Introduction to Robot Manipulation and Navigation, University of Texas at Dallas, Dallas, Texas, USA Fall 2024

CS 4391 Introduction to Computer Vision, University of Texas at Dallas, Dallas, Texas, USA Spring 2024

CS 6301 Special Topics in Computer Science: Introduction to Robot Manipulation and Navigation, University of Texas at Dallas, Dallas, Texas, USA Fall 2023

CS 6384 Computer Vision, University of Texas at Dallas, Dallas, Texas, USA Spring 2023

CS 6301 Special Topics in Computer Science: Introduction to Robot Manipulation and Navigation, University of Texas at Dallas, Dallas, Texas, USA Fall 2022

CS 6384 Computer Vision, University of Texas at Dallas, Dallas, Texas, USA Spring 2022

CS 6334 Virtual Reality, University of Texas at Dallas, Dallas, Texas, USA Fall 2021

Artificial Intelligence, University of Washington, Seattle, Washington, USA 2017
Guest Lectures for Prof. Dieter Fox

Computer Vision, University of Washington, Seattle, Washington, USA 2017
Guest Lecture for Prof. Linda Shapiro

Computer Vision, Stanford University, Stanford, California, USA 2016
Guest Lectures for Prof. Silvio Savarese

The C Programming Language, Fudan University, Shanghai, China Sep 2009 – Jan 2010
Teaching Assistant

STUDENT
MENTORSHIP

Current Ph.D. Students:

Jikai Wang, Ph.D. Student, University of Texas at Dallas	Aug 2021 – present
Jishnu Jaykumar Padalunkal, Ph.D. Student, University of Texas at Dallas	Aug 2021 – present
Qifan Zhang, Ph.D. Student, University of Texas at Dallas	Jan 2022 – present
Sai Haneesh Allu, Ph.D. Student, University of Texas at Dallas	Aug 2022 – present
Luis Felipe Casas Murillo, Ph.D. Student, University of Texas at Dallas	Aug 2023 – present
Jiahui Zhang, Ph.D. Student, University of Texas at Dallas	Aug 2025 – present

Current Master's Students:

Ahad Jawaidd, Master's Student, University of Texas at Dallas	Mar 2024 – present
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Current Undergraduate Students:

Jude Onyenze, Undergraduate Student, University of Texas at Dallas	Jan 2025 – present
Robert Teal, Undergraduate Student, University of Texas at Dallas	Jun 2025 – present
Keval Shah, Undergraduate Student, University of Texas at Dallas	Sep 2025 – present
Alex Huang, Undergraduate Student, University of Texas at Dallas	Sep 2025 – present
May Lynn Espinola, Undergraduate Student, University of Texas at Dallas	Sep 2026 – present

Current High School Students:

Daniel Yeh, Saratoga High School, California	May 2026 – present
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Past Ph.D. Students:

Rohith Peddi, Ph.D. Student, University of Texas at Dallas → Senior Machine Learning Engineer, LinkedIn	Aug 2022 – Aug 2026
Yangxiao Lu, Ph.D. Student, University of Texas at Dallas → Research Scientist, Meta	May 2022 – Oct 2025
Shivvrat Arya, Ph.D. Student, University of Texas at Dallas → Assistant Professor, NJIT	Aug 2021 – May 2025
Ninad Arun Khargonkar, Ph.D. Student, University of Texas at Dallas → Applied Scientist, Amazon	Nov 2021 – May 2025

Past Master's Students:

Saurav Nitin Dosi, Master's Student, University of Texas at Dallas	Feb 2024 – May 2026
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Animesh Maheshwari, Master's Student, University of Texas at Dallas	Feb 2024 – May 2025
Vinaya Bomnale, Master's Student, University of Texas at Dallas	Feb 2024 – May 2025
Peter Driscoll, Master's Student, University of Texas at Dallas	Feb 2024 – Dec 2024
Zhouhang Sun, Master's Student, University of Texas at Dallas	Feb 2024 – May 2024
Cole Salvato, Master's Student, University of Texas at Dallas	Aug 2021 – May 2024
Kamalesh Palanisamy, Master's Student, UT Dallas → ASML	Aug 2021 – Aug 2023
Abhishek Rajendra Prasad, Master's Student, UT Dallas → Goldman Sachs	Aug 2021 – Aug 2023
Priyanshi Shah, Master's Student, UT Dallas → Goldman Sachs	Feb 2022 – Aug 2023
Lirui Wang, Master's Student, University of Washington → Ph.D. student at MIT	Sep 2017 – Jul 2021

Past Undergraduate Students:

Itay Kadosh, Undergraduate Student, University of Texas at Dallas → Ph.D. student at University of Florida	Jul 2024 – May 2026
Govind Rangappa, Undergraduate Student, University of Texas at Dallas	Jan 2025 – Dec 2025
Michael Ugochukwu, Undergraduate Student, University of Texas at Dallas	Sep 2024 – Dec 2024
Nikhil Narvekar, Undergraduate Student, University of Texas at Dallas	Jan 2023 – Feb 2024
Jesse Musa, Undergraduate Student, University of Texas at Dallas	Sep 2023 – Feb 2024
Shaurya Dwivedi, Undergraduate Student, University of Texas at Dallas	Jan 2023 – Feb 2024
Hamzah Dweik, Undergraduate Student, University of Texas at Dallas	Jan 2023 – Feb 2024
Zesheng Xu, Undergraduate Student, University of Texas at Dallas	Aug 2021 – Feb 2024
Willie Chalmers III, Undergraduate Student, University of Texas at Dallas	May 2022 – Aug 2023
Gagan Bhat, Undergraduate Student, University of Texas at Dallas → Netflix	Jan 2022 – Dec 2022

Past High School Students:

May Lynn Espinola, Tompkins High School, Katy, TX → UT Dallas	May 2024 – May 2026
Gaurav Salandri, High School Student, Wakeland High School, Frisco → UT Austin	May 2022 – Feb 2024
Neil Song, High School Student, St. Mark's → Yale University	May 2021 – Aug 2023

NVIDIA Research Internship Program:

Xiangyun Meng, Ph.D. Student, University of Washington	2021
Luyang Zhu, Ph.D. Student, University of Washington	2020
Xinke Deng, Ph.D. Student, UIUC	2018, 2019
Keunhong Park, Ph.D. Student, University of Washington	2019
Corinne Jones, Ph.D. Student, University of Washington	2019
Raluca Scona, Ph.D. Student, Edinburgh Center for Robotics	2019
Fei Xia, Ph.D. Student, Stanford University	2018
Rui Wang, Ph.D. Student, TUM	2018
Christopher Xie, Ph.D. Student, University of Washington	2018

PROFESSIONAL SERVICE

Associate Editor

- IEEE Robotics and Automation Letters (RA-L) Oct 2021 – present
- International Conference on Robotics and Automation (ICRA) 2027
- International Conference on Robotics and Automation (ICRA) Workshops 2025, 2026
- International Conference on Intelligent Robots and Systems (IROS) 2022 – 2026

Area Chair

- Robotics: Science and Systems (RSS) 2024, 2025
- Conference on Robot Learning (CoRL) 2022, 2023, 2026

Publicity Chair

- Robotics: Science and Systems (RSS) 2026

Program Chair

- EgoAct: 1st Workshop on Egocentric Perception and Action for Robot Learning at RSS, 2025
- The Workshop for Neural Representation Learning for Robot Manipulation at CoRL, 2023
- The Visual Learning and Reasoning for Robotics Workshop at RSS, 2021
- The 3D Vision and Robotics Workshop at CVPR, 2021
- The Visual Learning and Reasoning for Robotic Manipulation Workshop at RSS, 2020
- 5th International IEEE Workshop on 3D Representation and Recognition at ICCV, 2015

Program Committee

- 4th International IEEE Workshop on 3D Representation and Recognition at ICCV, 2013

Tutorial Organizer

- 3D Object Geometry from Single Image Tutorial at International Conference on 3D Vision, 2016

Journal Reviewer

- International Journal of Robotics Research (IJRR)
- IEEE Robotics and Automation Letters (RA-L)
- IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)
- International Journal of Computer Vision (IJCV)
- Computer Vision and Image Understanding (CVIU)
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT)
- IEEE Transactions on Multimedia (TMM)
- IEEE Transactions on Signal Processing (TSP)

Conference Reviewer

- Robotics: Science and Systems Conference (RSS)
- IEEE International Conference on Robotics and Automation (ICRA)
- International Conference on Intelligent Robots and Systems (IROS)
- IEEE International Conference on Robot and Human Interactive Communication (RO-MAN)
- IEEE Conference on Computer Vision and Pattern Recognition (CVPR)
- International Conference on Computer Vision (ICCV)
- European Conference on Computer Vision (ECCV)
- Asian Conference on Computer Vision (ACCV)
- British Machine Vision Conference (BMVC)
- International Conference on 3D Vision (3DV)
- Neural Information Processing Systems (NeurIPS)

- INVITED TALKS
1. **From Modular Robotics Pipelines to Vision-Language-Action Systems: Lessons from Real-World Manipulation**
RIGOROUS Robot Perception at ICRA 2026, Vienna, Austria, 6/1/2026.
 2. **Cross-Embodiment Robotic Manipulation: Unifying Grippers Across Robots**
Texas Regional Robotics Symposium (TEROS), UT Arlington, 4/3/2026.
Robotics and AI Institute (RAI), Cambridge, Massachusetts, 10/30/2025.
BigAI, Brown University, Providence, Rhode Island, 10/31/2025.
 3. **Learning Robotic Manipulation from Human Demonstration Videos**
Guest Lecture, UNC-Chapel Hill (online), 10/15/2025.
Stanford Vision and Learning Lab, Stanford University, California, 5/19/2025.
 4. **Intelligent Robotics and Vision Lab**
MIT, Cambridge, Massachusetts, 4/4/2025.
 5. **Perceive, Plan and Act: Building Intelligent Robots in Human Environments**
Samsung Research America, Plano, Texas, 2/12/2025.
 6. **Building Intelligent Robots in Human Environments**
XPeng, Shenzhen, China, 6/6/2024.
 7. **Do We Need 3D Representations for Robot Manipulation?**
1st Workshop on 3D Visual Representations for Robot Manipulation, ICRA, Yokohama, Japan, 5/17/2024.
 8. **Connecting 6D Object Pose Estimation with Robot Manipulation**
8th International Workshop on Recovering 6D Object Pose, ICCV, Paris, France, 10/23/2023.
 9. **Object-Centric Perception for Robot Manipulation**
Texas Regional Robotics Symposium, Rice University, 4/14/2023.
 10. **Segmenting Unseen Objects for Robotic Grasping**
Guest Lecture, University of Minnesota, Twin Cities (online), 3/16/2023.
 11. **Closed-loop 6D Robotic Grasping of Unseen Objects**
Texas Regional Robotics Symposium, UT Austin, 4/29/2022.

12. **6D Robotic Grasping of Unseen Objects**
Electronics and Telecommunications Research Institute, South Korea, 8/19/2021.
13. **Perceive, Plan, Act and Learn: Towards Intelligent Robots in Human Environments**
UNC, 2/24/2021; UT Dallas, 3/16/2021.
14. **Learning RGB-D Feature Embeddings for Unseen Object Instance Segmentation**
NVIDIA Research, Seattle, Washington, 10/12/2020.
15. **PoseRBPF: A Rao-Blackwellized Particle Filter for 6D Object Pose Tracking**
University of Washington, Seattle, Washington, 9/27/2019.
16. **Object Perception for Robot Manipulation**
Toyota Research Institute, Cambridge, Massachusetts, 7/12/2019.
17. **Perceiving the 3D World from Images and Videos**
NVIDIA Research, Redmond, Washington, 11/07/2017; University of Michigan, 3/15/2018.
18. **3D Object Recognition and Scene Understanding from RGB-D Videos**
GRASP Lab at University of Pennsylvania, 10/11/2017; Microsoft Research, Redmond, 10/17/2017; Vision Lab at Stanford University, 10/23/2017.
19. **3D Object Recognition and Scene Understanding**
Mitsubishi Electric Research Laboratories, Boston, Massachusetts, 7/14/2017.
20. **Tutorial on 3D Object Recognition**
International Conference on 3D Vision, Stanford University, 10/28/2016.
21. **3D Object Representations for Recognition**
Carnegie Mellon University, 3/28/2016; University of Toronto, 4/4/2016; Massachusetts Institute of Technology, 4/12/2016; University of California, Berkeley, 4/21/2016; University of Illinois at Urbana-Champaign, 5/5/2016; University of Washington, 5/31/2016.
22. **3D Object Detection and Pose Estimation**
The 1st International Workshop on Recovering 6D Object Pose in conjunction with ICCV, Santiago, Chile, 12/17/2015.

- CONFERENCE PRESENTATIONS
1. **From Local Matches to Global Masks: Template-Guided Instance Detection and Segmentation in Open-World Scenes**
In Robotics: Science and Systems (RSS), Sydney, Australia, 7/16/2026.
 2. **PoseCNN: A Convolutional Neural Network for 6D Object Pose Estimation in Cluttered Scenes**
In Robotics: Science and Systems (RSS), CMU, Pittsburgh, Pennsylvania, 6/26/2018.
 3. **DA-RNN: Semantic Mapping with Data Associated Recurrent Neural Networks**
In Robotics: Science and Systems (RSS), Massachusetts Institute of Technology, Massachusetts, 7/13/2017.
 4. **Subcategory-aware Convolutional Neural Networks for Object Proposals and Detection**
In IEEE Winter Conference on Applications of Computer Vision, Santa Rosa, California, 3/29/2017.
 5. **Learning to Track: Online Multi-Object Tracking by Decision Making**
In International Conference on Computer Vision, Santiago, Chile, 12/16/2015.
 6. **Data-Driven 3D Voxel Patterns for Object Category Recognition**
In IEEE Conference on Computer Vision and Pattern Recognition, Boston, Massachusetts, 6/8/2015.
 7. **Monocular Multiview Object Tracking with 3D Aspect Parts**
In the 1st Stanford-SNU Workshop on Automated Driving, Stanford University, 2/24/2015.
 8. **Beyond PASCAL: A Benchmark for 3D Object Detection in the Wild**
In IEEE Winter Conference on Applications of Computer Vision, Steamboat Springs, Colorado, 3/24/2014.
 9. **Object Detection by 3D Aspectlets and Occlusion Reasoning**
In the 4th International IEEE Workshop on 3D Representation and Recognition in conjunction with ICCV, Sydney, Australia, 12/8/2013.
 10. **Estimating the Aspect Layout of Object Categories**
In Midwest Vision Workshop, University of Illinois at Urbana-Champaign, 9/21/2012.